

# WRRRA-M 0.2 Cross Dimensional Filter Correspondence

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Conditional Sixteen Channel Reorganization Hypercharge and Anomaly Closure

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## Research decision

This note closes one question left open in WRRRA-M 0.1. It asks whether the sixteen channels can be reorganized by a single cross-dimensional filter so that weak pairing, hypercharge, electric charge, and anomaly cancellation emerge together. The answer is conditional and exact within the stated assumptions.

## Abstract

WRRRA-M 0.2 defines a sixteen-channel space consisting of one invariant channel, two seven-dimensional sectors, and one null channel. It then introduces a conditional cross-dimensional filter that reorganizes the space into the Pati-Salam representation  $(4,2,1)$  plus  $(\text{anti-}4,1,2)$ . The two seven sectors supply paired color triplets and paired antitriplets, while their singlets and the two distinguished one-dimensional channels complete the corresponding fours. Color tracelessness fixes the  $B$  minus  $L$  generator. Neutrality of the null channel and unit normalization of the invariant channel then fix the unique hypercharge relation  $Y$  equals  $T3R$  plus one half  $B$  minus  $L$ . The Standard Model hypercharges, electric charges, perturbative gauge anomalies, mixed gravitational anomaly, and global  $SU(2)$  parity check follow. The result does not derive the existence or dynamics of the filter and does not claim that  $G2$  alone generates the full Pati-Salam group.

## Result in one sentence

WRRRA-M 0.2 does not prove that the cross-dimensional filter exists. It proves that if such a filter exists with the stated channel pairing and two normalization conditions, the one-generation sixteen-channel placement, hypercharge, electric charge, and anomaly cancellation close together.

## Scope and stopping line

The target is deliberately narrow. This version neither extends the cosmological architecture nor explains why nature selects this filter. It completes the algebraic step that 0.1 identified as the next decisive test and then stops.

### Included in 0.2

- Definition of the sixteen-channel input space
- Definition of the conditional cross-dimensional filter
- Exact channel counting and Pati-Salam correspondence
- Conditional derivation of hypercharge and electric charge
- Exact anomaly checks and explicit falsification conditions

### Reserved for 0.3 or later

- Filter dynamics or a derivation from Source and Law
- Three generations, masses, Higgs structure, or neutrino mass
- Gravity, dark matter, cosmology, or consciousness
- A claim that  $G2$  by itself generates the full Pati-Salam group

## External mathematical baseline

The Pati-Salam representation and the sixteen-component spinor organization associated with SO(10) are established external structures, not WRRR discoveries [1][2]. The G2 branching 7 to 3 plus anti-3 plus 1 is also an external representation-theoretic input [3]. WRRR-M contributes a particular conditional reorganization of its one plus seven plus seven plus one-null channel space into that known structure.

## Fixed WRRR evaluation rule

| Stage                        | Question answered in this note                                              |
|------------------------------|-----------------------------------------------------------------------------|
| Verification input           | Which established representation identities and anomaly conditions are used |
| WRRR-specific transformation | How the cross filter rearranges the four input sectors                      |
| Output                       | Which charges and consistency relations follow from the rearrangement       |
| Falsification condition      | Which failure would reject this 0.2 correspondence                          |

## The sixteen-channel input

The 0.2 input adds the WRRR null channel to the fifteen-channel space of 0.1. The subscripts distinguish channel origins; they do not yet denote observed particles. The invariant channel 1-zero and null channel 1-N are structurally distinct one-dimensional sectors.

$$C_{16} = \mathbf{1}_0 \oplus \mathbf{7}_A \oplus \mathbf{7}_B \oplus \mathbf{1}_N$$

Compact G2 has a real seven-dimensional fundamental representation. Here each seven denotes its complexification, whose SU(3) restriction is 3 plus anti-3 plus 1. The full input has two triplets, two antitriplets, and four singlets, totaling sixteen complex color components. Reading these as sixteen independent complex left-handed Weyl channels is an additional physical assumption: the triplet and antitriplet fields are not identified by the original real-representation conjugation. The color count alone does not determine chirality [5].

$$\mathbf{7} \downarrow_{SU(3)} = \mathbf{3} \oplus \bar{\mathbf{3}} \oplus \mathbf{1}$$

| Input sector | Color decomposition  | Channel count | Role after filtering      |
|--------------|----------------------|---------------|---------------------------|
| 1-zero       | 1                    | 1             | Visible invariant channel |
| 7-A          | 3 plus anti-3 plus 1 | 7             | First paired components   |

| Input sector | Color decomposition  | Channel count | Role after filtering     |
|--------------|----------------------|---------------|--------------------------|
| 7-B          | 3 plus anti-3 plus 1 | 7             | Second paired components |
| 1-N          | 1                    | 1             | Neutral null channel     |

## The cross-dimensional filter

Define DF-cross as a conditional linear reorganization that preserves the SU(3) color action while pairing channels across the two seven sectors. It does not create dimensions or particles. It changes which input components are read together as phenotype multiplets.

$$DF_{\times} : C_{16} \longrightarrow (\mathbf{4}, \mathbf{2}, \mathbf{1}) \oplus (\bar{\mathbf{4}}, \mathbf{1}, \mathbf{2})$$

The triplet and singlet from sector A are paired with their counterparts from sector B as the two components of an SU(2)L doublet. Their combined dimension is eight.

$$\mathcal{H}_L = (\mathbf{3}_A \oplus \mathbf{1}_A) \oplus (\mathbf{3}_B \oplus \mathbf{1}_B) \cong (\mathbf{4}, \mathbf{2}, \mathbf{1})$$

The anti-triplet from sector A is paired with the null singlet, and the anti-triplet from sector B with the invariant singlet. These two anti-fours form an SU(2)R doublet, again of dimension eight.

$$\mathcal{H}_R = (\bar{\mathbf{3}}_A \oplus \mathbf{1}_N) \oplus (\bar{\mathbf{3}}_B \oplus \mathbf{1}_0) \cong (\bar{\mathbf{4}}, \mathbf{1}, \mathbf{2})$$

| Filtered block | Channel assignment                             | Dimension               | External notation |
|----------------|------------------------------------------------|-------------------------|-------------------|
| Left block     | (3-A plus 1-A) and (3-B plus 1-B)              | 4 times 2 equals 8      | (4,2,1)           |
| Right block    | (anti-3-A plus 1-N) and (anti-3-B plus 1-zero) | anti-4 times 2 equals 8 | (anti-4,1,2)      |

## Conditional correspondence theorem

Assume the independent complex left-handed Weyl-channel assignment above. Assume that DF-cross exists, preserves the common SU(3) action, and produces the two blocks above. Assume also that 1-N is neutral and 1-zero has unit hypercharge. Then the sixteen input channels carry exactly one left-handed Standard Model generation including the conjugate right-handed neutrino, with the standard hypercharges and electric charges. The listed gauge and mixed anomalies vanish.

## Proof by channel reorganization

The left block contains two color triplets and two color singlets. Reading A and B as the SU(2)L pair gives the quark doublet Q-L and lepton doublet L-L. The right block contains two anti-triplets and two singlets. Reading the A and B groupings as the SU(2)R pair gives (u-c, nu-c) at T3R minus one half and (d-c, e-c) at T3R plus one half. No channel is added or discarded.

## Hypercharge from two normalization conditions

Within an SU(4) four, color symmetry gives equal entries for the three color components. Tracelessness then fixes the lepton entry relative to them. Choosing the conventional baryon-minus-lepton normalization gives the diagonal generator below; its signs reverse on the anti-four.

$$B - L = \text{diag}\left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}, -1\right)_4, \quad - (B - L)_4 \text{ on } \bar{4}$$

Take the most general linear combination of the two commuting diagonal generators relevant after the filter.

$$Y = \alpha T_{3R} + \beta (B - L)$$

The null channel is assigned to nu-c with T3R minus one half and B minus L equal to plus one. Its required neutrality gives beta equals alpha over two. The invariant channel is assigned to e-c with T3R plus one half and B minus L equal to plus one. Unit hypercharge then gives alpha equals one. Thus a single operator, rather than six separate inserted values, fixes all channel charges.

$$0 = Y(\nu^c) = -\frac{\alpha}{2} + \beta, \quad 1 = Y(e^c) = \frac{\alpha}{2} + \beta \implies \alpha = 1, \beta = \frac{1}{2}$$

$$Y = T_{3R} + \frac{B-L}{2}$$

## Recovered channel charges

The left block is an SU(2)R singlet, so T3R is zero there. The right block uses the two eigenvalues plus or minus one half. Applying the derived operator gives:

$$Q_{\text{electric}} = T_{3L} + Y$$

| Field | Multiplicity | B minus L | T3R       | Y         | Electric charges  |
|-------|--------------|-----------|-----------|-----------|-------------------|
| Q-L   | 6            | 1/3       | 0         | 1/6       | 2/3 and minus 1/3 |
| L-L   | 2            | minus 1   | 0         | minus 1/2 | 0 and minus 1     |
| u-c   | 3            | minus 1/3 | minus 1/2 | minus 2/3 | minus 2/3         |
| d-c   | 3            | minus 1/3 | plus 1/2  | 1/3       | 1/3               |
| nu-c  | 1            | 1         | minus 1/2 | 0         | 0                 |

| Field | Multiplicity | B minus L | T3R      | Y | Electric charges |
|-------|--------------|-----------|----------|---|------------------|
| e-c   | 1            | 1         | plus 1/2 | 1 | 1                |

## Exact consistency checks

All sums below are over the declared left-handed Weyl fields. The standalone script `calculations/verify_wrra_m_0_2.py` solves the two normalization conditions, applies the single charge operator to all sixteen channels, and checks electric charges, the listed local anomaly sums, and both SU(2) doublet parities with exact rational arithmetic.

$$\sum Y = 6\left(\frac{1}{6}\right) + 3\left(-\frac{2}{3}\right) + 3\left(\frac{1}{3}\right) + 2\left(-\frac{1}{2}\right) + 1 + 0 = 0$$

$$\sum Y^3 = 6\left(\frac{1}{6}\right)^3 + 3\left(-\frac{2}{3}\right)^3 + 3\left(\frac{1}{3}\right)^3 + 2\left(-\frac{1}{2}\right)^3 + 1 = 0$$

$$A_{SU(3)^2U(1)} = 2\left(\frac{1}{2}\right)\left(\frac{1}{6}\right) + \frac{1}{2}\left(-\frac{2}{3}\right) + \frac{1}{2}\left(\frac{1}{3}\right) = 0$$

$$A_{SU(2)^2U(1)} = 3\left(\frac{1}{2}\right)\left(\frac{1}{6}\right) + \frac{1}{2}\left(-\frac{1}{2}\right) = 0$$

| Condition            | Channel calculation                                            | Result          |
|----------------------|----------------------------------------------------------------|-----------------|
| SU(3) cubed          | Two triplets minus two antitriplets                            | 0               |
| SU(3) squared U(1)   | Q-L plus u-c plus d-c                                          | 0               |
| SU(2)L squared U(1)  | Three colored Q-L doublets plus L-L                            | 0               |
| U(1) cubed           | Six weighted hypercharge groups                                | 0               |
| Gravity squared U(1) | Sum of all sixteen hypercharges                                | 0               |
| Global SU(2)L        | Three colored quark doublets plus one lepton doublet           | 4 doublets even |
| Global SU(2)R        | Three colored anti-quark doublets plus one anti-lepton doublet | 4 doublets even |

The even-doublet condition is the global SU(2) consistency test identified by Witten [4]. The same parity count holds independently for the left and right SU(2) factors.

## Claim ledger

| Code | Status                  | Statement in WRR-M 0.2                                                                             |
|------|-------------------------|----------------------------------------------------------------------------------------------------|
| E    | External input          | Pati-Salam multiplets, $SO(10)$ sixteen structure, G2 branching, and anomaly criteria              |
| D    | Architecture definition | The input C16 and the conditional target of DF-cross                                               |
| C    | Conditional calculation | Exact rearrangement, unique normalized hypercharge operator, electric charges, and anomaly closure |
| H    | Unproved hypothesis     | Existence and dynamical selection of DF-cross from Source and Law                                  |
| O    | Outside this version    | Generations, masses, symmetry breaking, gravity, dark matter, and consciousness                    |

## Falsification conditions

This 0.2 correspondence fails if any one of the following is established:

- No linear filter can implement the stated pairing while preserving the common  $SU(3)$  action.
- The four input sectors cannot be kept independent without duplicating or deleting a channel.
- Neutrality of 1-N and unit normalization of 1-zero do not yield one operator valid for every channel.
- Any listed local anomaly sum is nonzero or either  $SU(2)$  factor contains an odd number of doublets.
- A later dynamical model requires a filter incompatible with the 0.2 pairing.

## The one remaining assumption

The independent complex Weyl-channel and chirality assignments are declared physical inputs. Given those inputs, the remaining filter question is why Source and Law realize DF-cross. This note defines its required action and proves its algebraic consequences; existence, uniqueness, stability, and physical selection of the filter remain open.

## Conclusion

WRR-M 0.1 ended with an exact color-sector count and an unresolved request for weak pairing and one hypercharge operator. Version 0.2 supplies a conditional answer. By crossing components of the two seven sectors with the invariant and null singlets, the sixteen channels reorganize into

a known left-right structure. Two transparent normalization conditions then select one hypercharge operator, and all one-generation charges and anomaly checks close without channel-by-channel fitting. The correspondence is therefore stronger than numerical resemblance but narrower than a fundamental derivation. Its unfinished part is stated plainly: why the universe realizes this filter at all.

## Appendix A Compact derivation

| Step | Input                        | WRRM-M operation                                 | Output                                      |
|------|------------------------------|--------------------------------------------------|---------------------------------------------|
| 1    | 1 plus 7-A plus 7-B plus 1-N | Apply SU(3) branching                            | Two 3, two anti-3, four 1                   |
| 2    | Sixteen color channels       | Cross-pair A, B, invariant, and null sectors     | (4,2,1) plus (anti-4,1,2)                   |
| 3    | T3R and B minus L            | Require $Y(\text{nu-c})=0$ and $Y(\text{e-c})=1$ | Y equals T3R plus (B minus L)/2             |
| 4    | One derived operator         | Apply to all channels                            | Six Standard Model field-group hypercharges |
| 5    | Derived hypercharges         | Evaluate exact anomaly sums                      | All listed consistency tests pass           |

$$C_{16} = \mathbf{1}_0 \oplus \mathbf{7}_A \oplus \mathbf{7}_B \oplus \mathbf{1}_N$$

$$DF_x : C_{16} \longrightarrow (\mathbf{4}, \mathbf{2}, \mathbf{1}) \oplus (\overline{\mathbf{4}}, \mathbf{1}, \mathbf{2})$$

$$Y = T_{3R} + \frac{B-L}{2}$$

## Appendix B References

- [1] Jogesh C. Pati and Abdus Salam Lepton Number as the Fourth Color Physical Review D 10 1974 275 to 289  
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